



NITI Aayog

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Sector at a glance

Manufacturing

An Overview

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Sector at a glance

Manufacturing

Manufacturing in India

The **‘Sector-at-a-Glance’** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector

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Overview



Development of the manufacturing sector is critical from the point of view of ensuring sustainable growth by providing value addition to India's natural and agricultural resources, addressing strategic needs, creating employment opportunities for a large population, and developing new technologies for the welfare of citizens .

India's manufacturing sector plays a significant role in the country's economy, accounting for **13.3%** of the **GDP** as of 2022¹

The manufacturing sector is a major source of employment in India, providing jobs to **27.3** million people. This underscores its importance as a vital economic engine, contributing to India's overall growth and development.²

Every job created in manufacturing has a multiplier effect of creating **two to three additional jobs** in related activities¹.

India aims to achieve a robust economic trajectory, setting a GDP target of over by 2047. This strategic vision underscores the nation's consistent progression towards a developed nation. The manufacturing sector will play an instrumental role in India's pursuit to achieve this target. Beyond its contribution to economic growth, manufacturing significantly influences job creation and innovation, with each manufacturing job creating a ripple effect of two to three additional jobs in related activities.

As of 2022, the manufacturing sector accounted for \$16.3 trillion of global Gross Domestic Product (GDP), with China, the United States, and Germany collectively accounting for approximately 52% of the world's total manufacturing GDP. China holds the leading position, contributing around 30% to the total manufacturing GDP, driven by proactive government initiatives and favorable policies supporting the manufacturing sector and its role in global value chains. Despite the United States' shift towards a service-oriented economy, it maintains a 17% share in total manufacturing GDP, largely due to its status as the largest global economy and substantial reliance on outsourcing. Germany, recognized for its technological capabilities, consistently maintains a 5% share in the overall global manufacturing GDP.

The sustained significance of China, the United States, and Germany as key manufacturing destinations can be attributed to their economic strengths, including the strategic leveraging of economic size, government initiatives, and technological capabilities. India has recorded a Compound Annual Growth Rate (CAGR) of approximately 4% between 2011 and 2022, resulting in a 3% share of the global manufacturing GDP and signalling its evolving role in the global manufacturing landscape.

The importance of developing a comprehensive manufacturing ecosystem cannot be overstated. One primary objective is to curtail import dependency, thereby fortifying self-sufficiency and mitigating economic vulnerabilities. A strong manufacturing sector is not only an economic driver but a strategic lever to boost exports, contributing substantively to foreign exchange earnings and fortifying India's position in global value chain. A well-rounded manufacturing ecosystem

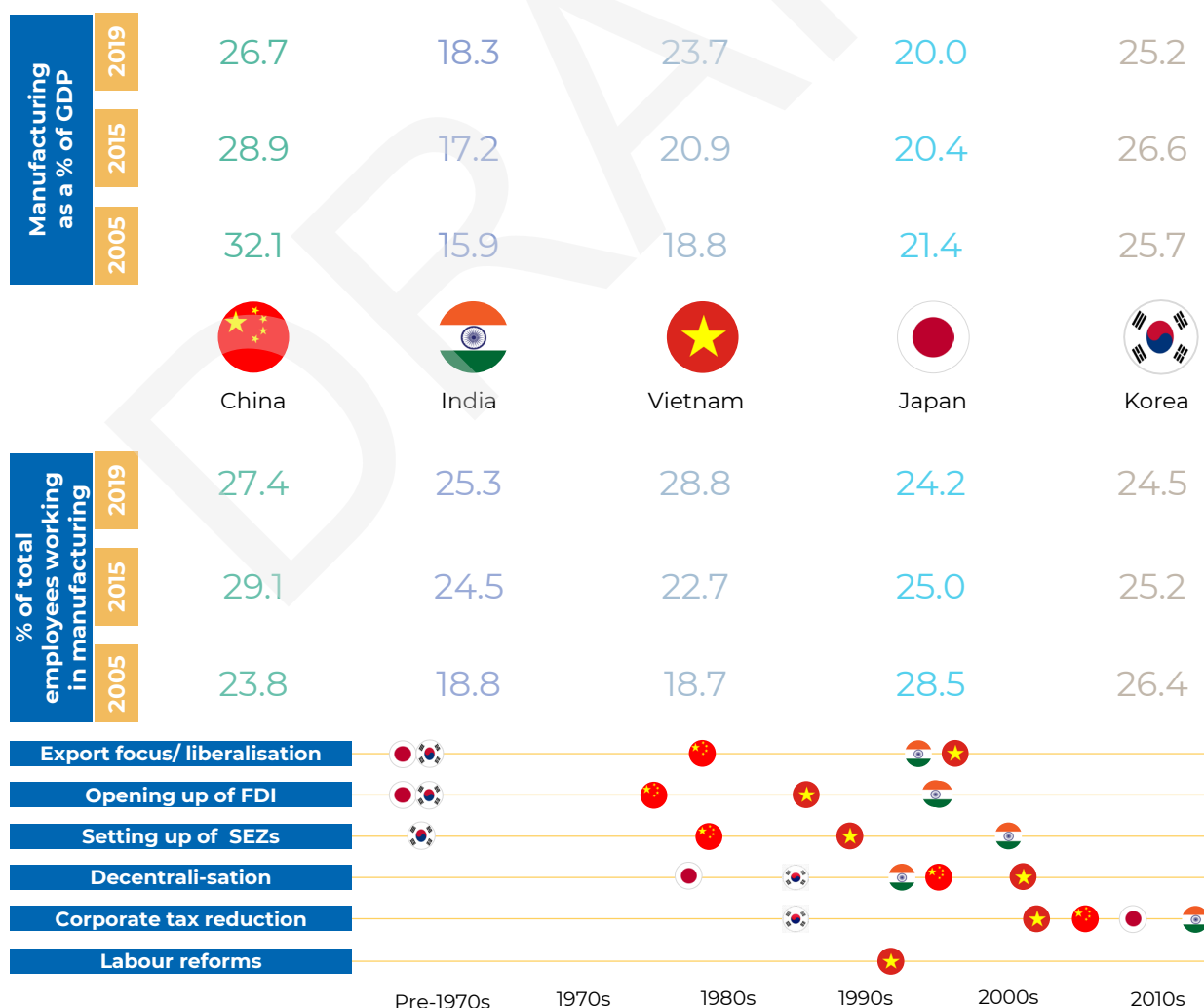
¹ World Bank

² Trade Promotion Council of India

aligns with a broader economic strategy, fostering resilience and sustainability. By nurturing innovation and employment opportunities, a thriving manufacturing sector acts as a catalyst for overall economic development. It serves as a linchpin in the pursuit of a self-reliant and globally competitive economy, aligning with India's overarching goal of securing a formidable position on the global economic stage by 2047.

Manufacturing has been pivotal in transforming the United States, Europe, and Japan into advanced economies. In the U.S., post-Industrial Revolution policies focused on research, innovation, and infrastructure, securing its global manufacturing dominance. Japan prioritized manufacturing during post-war reconstruction, focusing on quality and efficiency, bringing up its global competitiveness. In Europe, collaborative economic initiatives, such as the European Economic Community, streamlined trade and manufacturing growth. Germany's expertise in innovative manufacturing positioned Europe as a global player. These nations not only excelled domestically but also became integral in the global value chain, leveraging their advanced manufacturing capabilities to influence and shape international trade dynamics, reinforcing their positions as key players in the global economic landscape.

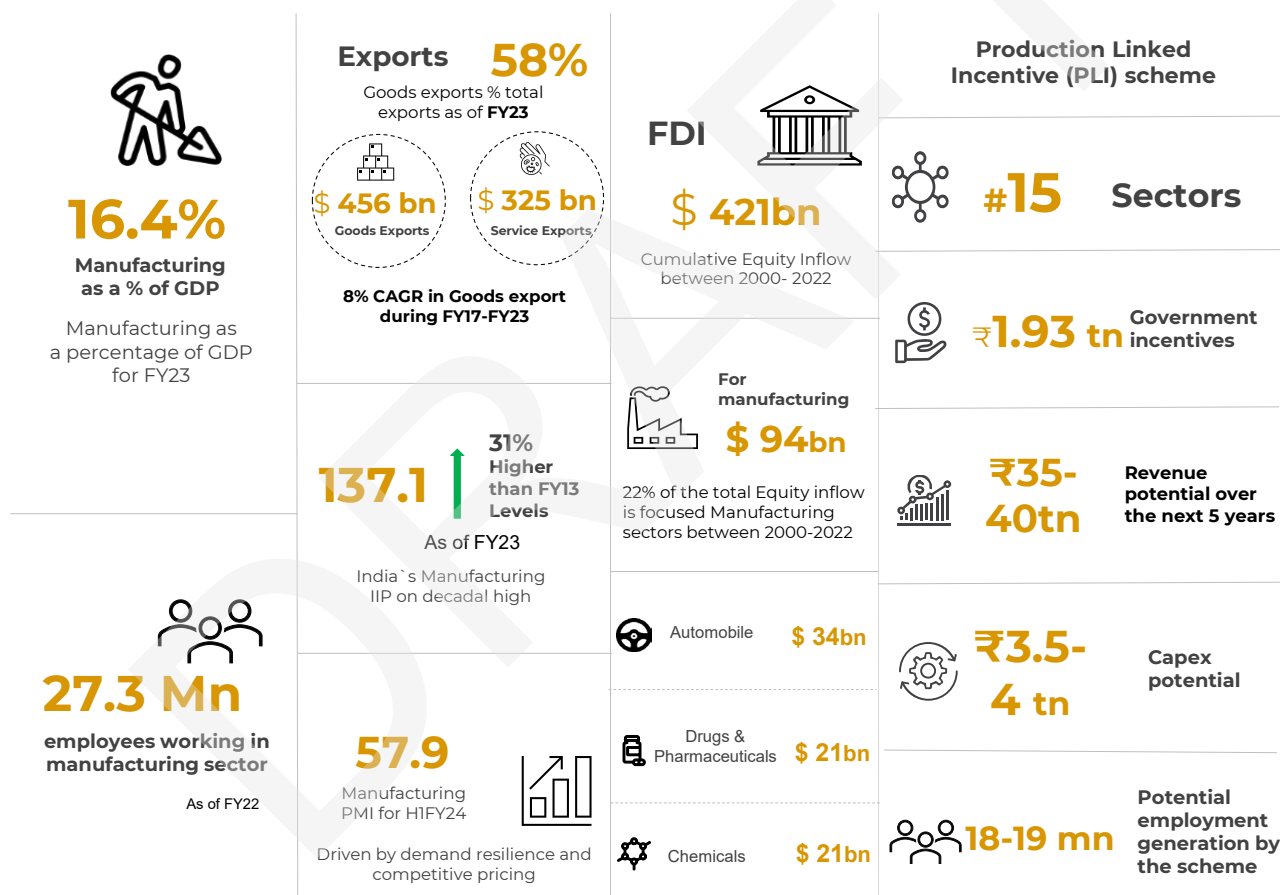
Figure 1: Benchmarking India with peers on Manufacturing and policy reforms



Source: Ministry of Statistics And Programme Implementation, World Bank
Manufacturing in India

China stands out with the highest share of manufacturing in its GDP among Asian economies, reaching 26.7% in 2019. This growth is credited to strategic policies such as trade liberalization and the establishment of special economic zones (SEZs), which elevated incomes and positioned China as a major global exporter. The Manufacturing GDP in China demonstrated consistent growth at a 6% rate from 2011 to 2019, reaching \$3.8 trillion. This underscores the effectiveness of China's policies in shaping a robust manufacturing landscape and contributing to overall economic growth. Similarly, Vietnam has experienced notable progress in manufacturing, propelled by government initiatives like corporate tax reduction. This strategic move facilitated a healthy Compound Annual Growth Rate (CAGR) of 12% in the manufacturing GDP between 2011 and 2019, showcasing significant sectoral advancements.

Figure 2: India's Manufacturing sector at a glance

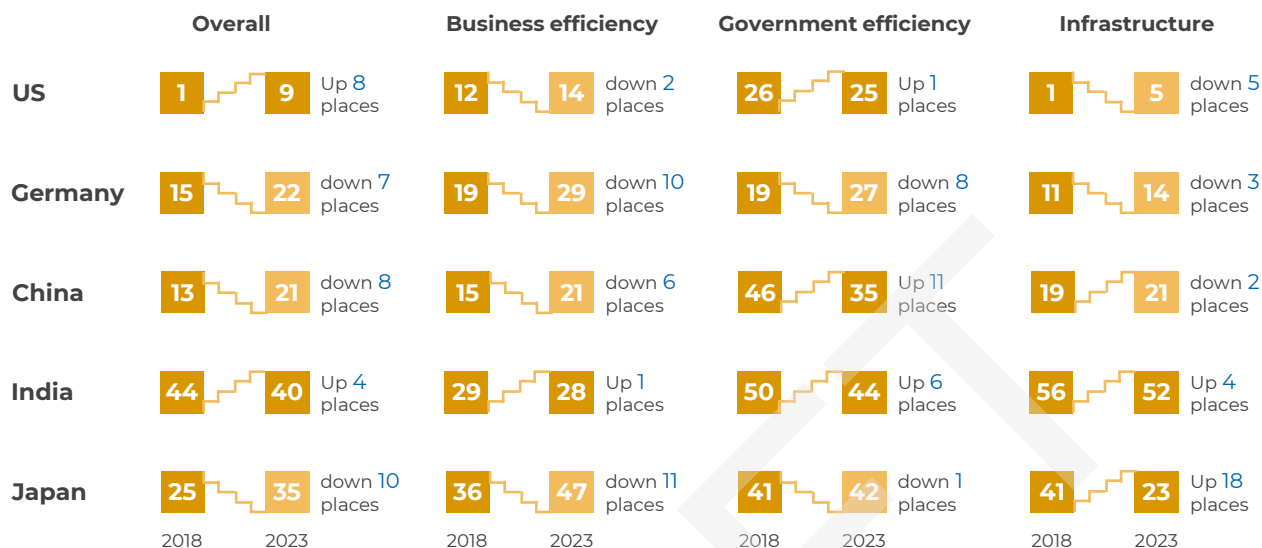


Source: 1. Reserve Bank of India
2. Department for Promotion of Industry and Internal Trade
3. Ministry of Statistics and Programme Implementation

Global Competitive Landscape: Stable developed nations and India's upward ascent

The World Competitiveness Ranking by IMD (International Institute for Management Development) is based on a comprehensive assessment of factors that contribute to a country's competitiveness. These factors include economic performance, government efficiency, business efficiency, and infrastructure, among others. The ranking provides insights into a nation's ability to foster an environment conducive to business competitiveness and sustainable economic growth.

Figure 3: World Competitiveness Ranking



Source: International Institute for Management Development (IMD)

In the current global competitive landscape, the United States, Germany, and Japan have maintained their stable standings, reflecting their well-established economic strengths and consistent policy frameworks. China's ascent underlines its sustained growth trajectory and relentless innovation and policy push, bolstering its competitiveness on the global stage.

In contrast, India has witnessed a positive transformation in its ranking, climbing from 44 to 37 over the past 4 years. This upward mobility underscores the country's progressive reforms and pro-business initiatives, which have attracted increased investments and created a more conducive environment for economic growth.

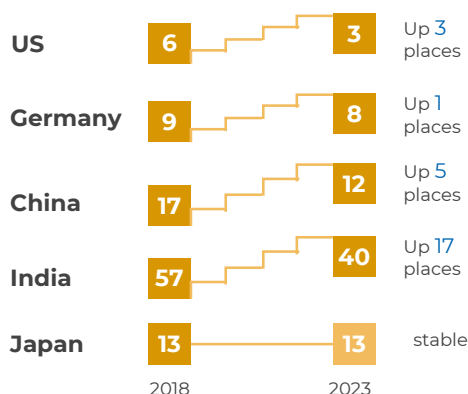
Global Innovation Index: Shifting Landscapes and Investment Impact

The Global Innovation Index (GII) is based on an assessment of a country's innovation capabilities and performance. It is published by the World Intellectual Property Organization (WIPO), Cornell University, and INSEAD. The index considers various indicators related to innovation, including:

1. **Institutions:** The political and regulatory environment that supports innovation.
2. **Human capital and research:** Factors such as education, research and development (R&D) activities, and the availability of skilled labor.
3. **Infrastructure:** The quality of infrastructure, including information and communication technology.
4. **Market and business sophistication:** The efficiency and sophistication of markets, and the level of business collaboration in innovation.
5. **Business environment:** The overall business climate, including the ease of starting a business and the level of competition.

The GI provides a comprehensive view of a country's innovation landscape, helping policymakers and businesses understand and promote innovation-driven growth.

Figure 4: Global innovation index



Source: World Intellectual Property Organization

India's ascent on the Global Innovation Index, rising from the 57th position in 2018 to 40th place in 2023 underscores India's growing innovation ecosystem, policy direction that spurs innovation, characterized by increased investments in research and development, a surge in patent applications, and a thriving start-up culture. The nation's commitment to fostering innovation and technological progress has propelled it onto the global innovation stage, signalling a promising future for India's creative and inventive potential. India ranks first in Central and Southern Asia and in Lower middle-income groupings on the index. However, while India has made significant strides in innovation, it still lags some of its global peers, highlighting the need for continued efforts to enhance its innovation ecosystem further.

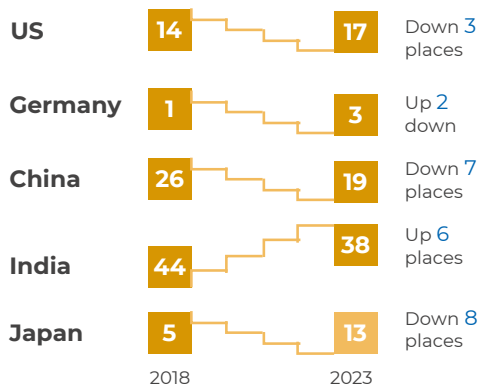
Logistics Index: Maximizing Potential through Logistics Excellence

The Logistics Performance Index (LPI) is based on a set of indicators that assess the efficiency and effectiveness of a country's logistics and supply chain infrastructure. It is published by the World Bank and is derived from a combination of quantitative data and qualitative assessments gathered through surveys.

Key components of the Logistics Performance Index include:

- 1. Customs and border management:** This assesses the efficiency of customs clearance processes and border management procedures.
- 2. Infrastructure quality:** It evaluates the quality of transport infrastructure, including roads, ports, and airports.
- 3. Ease of arranging shipments:** This measures the simplicity and efficiency of arranging shipments, including the quality of logistics services.
- 4. Competence and quality of logistics services:** This assesses the competence and quality of logistics services, including transport operators and customs brokers.
- 5. Tracking and tracing:** This evaluates the ability to track and trace consignments during transportation.
- 6. Timeliness of shipments:** It measures the timeliness of shipments in reaching their destination.
- 7. Frequency with which shipments reach consignees within scheduled or expected delivery times:** This assesses the reliability of delivery timelines.

Figure 5: Logistics performance index



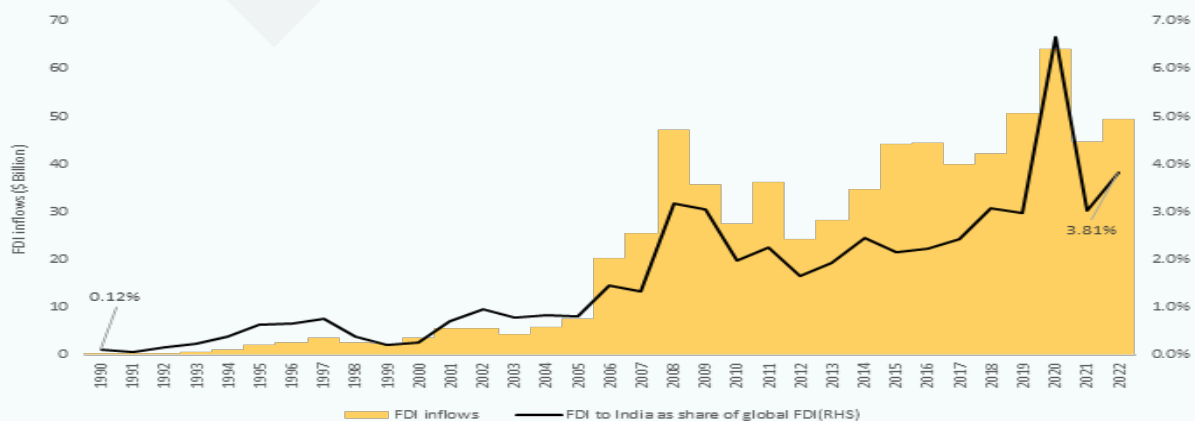
Source: World Bank

India's ascent in the Logistics Performance Index (LPI) ranking, advancing from 44th in 2018 to 38th in 2023, is a testament to strategic initiatives. Programs like Bharatmala, Sagarmala, and the infrastructure expansion through the National Infrastructure Pipeline, along with the GatiShakti Initiative, have played pivotal roles. The launch of the National Logistics Policy in 2022 underscores the acknowledgment of logistics challenges in India and a dedicated focus on improvement. These concerted efforts showcase India's commitment to optimizing its logistics ecosystem, a crucial step in bolstering its global competitiveness.

India a preferred FDI destination

India has implemented several measures to enhance Foreign Direct Investment (FDI) inflows, recognizing the pivotal role it plays in fostering economic growth and development. Key reforms include the liberalization of FDI norms across various sectors, simplification of procedures, and initiatives like 'Make in India' and 'Invest India' to attract foreign investors. The introduction of a transparent and predictable policy environment, coupled with efforts to improve the ease of doing business, has significantly contributed to an increase in FDI inflows enabling India to emerge as a preferred investment destination among global investors. FDI inflows to India as a percentage of the world FDI have improved from 0.12% in the pre-liberalisation era of 1990 to 3.81% in 2022.

Figure 6: Trend of FDI investments in India

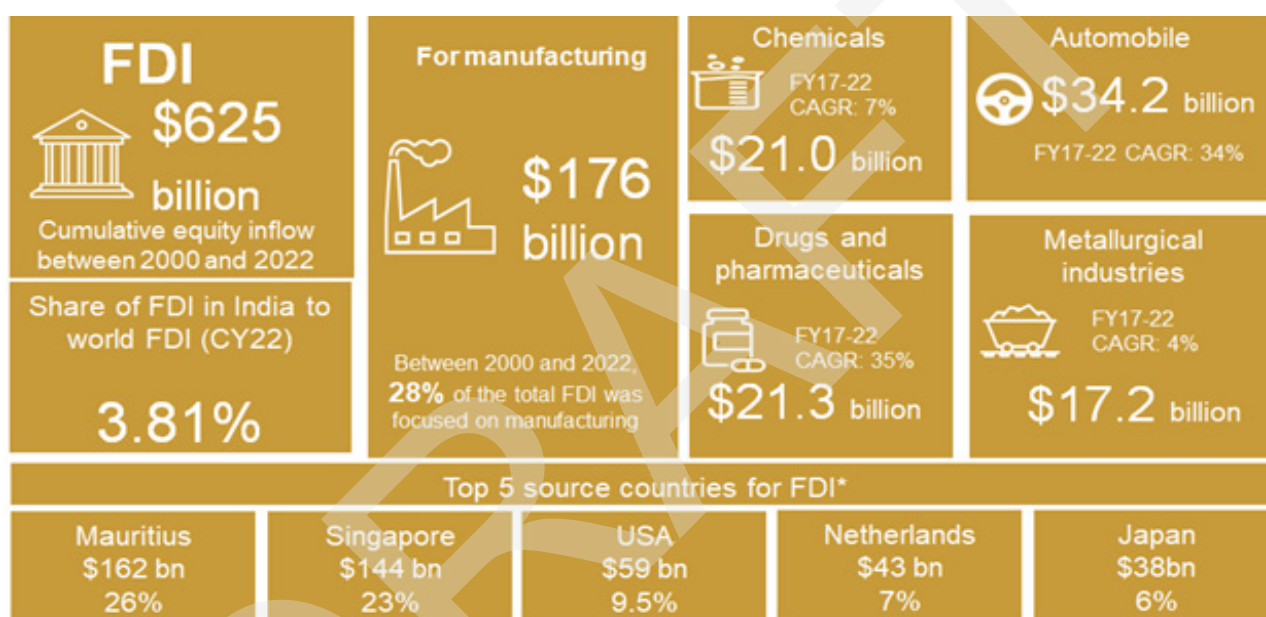


Note: FDI only includes Equity as defined by the United Nations conference on trade and development

Source: UNCTAD

Foreign direct investment brings not only financial capital but also technology, managerial expertise, and access to global markets. This infusion of foreign capital facilitates the development of critical infrastructure, boosts industrial growth, and stimulates job creation. Additionally, FDI contributes to the transfer of skills and knowledge, fostering innovation and improving the overall competitiveness of domestic industries. It serves as a catalyst for economic reforms, helps balance trade deficits, and strengthens India's integration into the global economy. As a result, a robust and sustained influx of FDI is essential for achieving sustainable economic development and realizing India's full growth potential on the world stage.

Figure 7: Snapshot of FDI investments in India



Note: * Date for FDI is cumulative from April 2020 to December 2022

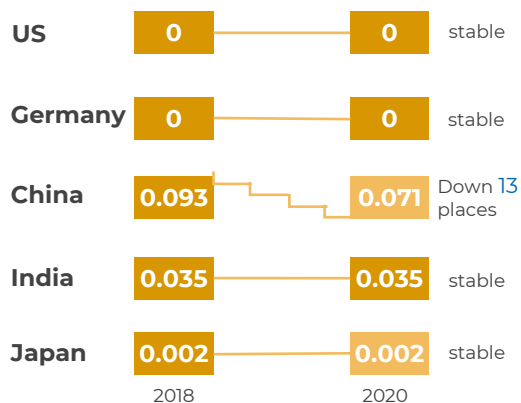
Percentage is the share of the particular country in overall FDI received over April 2020 to December 2022

Source: Ministry of Statistics and Programme Implementation, RBI

Nations Navigate Foreign Investment Regulation for Economic Growth

The FDI Restrictiveness Index measures the degree of restrictions imposed by a country on foreign direct investment (FDI). It quantifies regulatory barriers and limitations, providing insights into a nation's openness to foreign capital. A higher index value indicates greater restrictions, potentially discouraging foreign investors, while a lower index reflects a more liberalized environment conducive to international investment. Policymakers use this index to assess and compare the attractiveness of different countries for FDI.

Figure 8: FDI restrictiveness Index



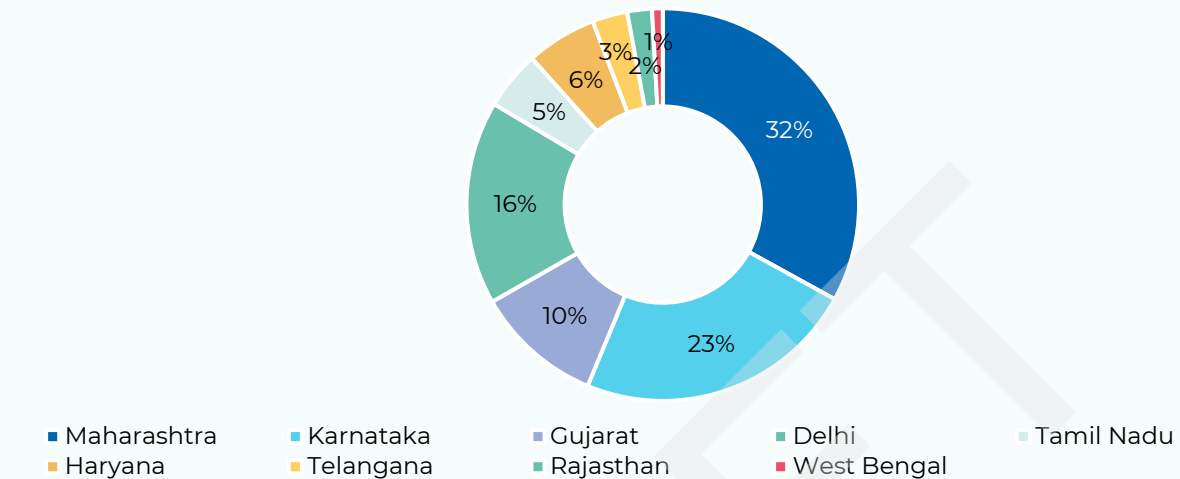
Source: OECD

Examining data spanning 2018 to 2020 reveals that prominent nations such as Germany, the United States, and Japan exhibit either no or minimal restrictions on Foreign Direct Investment (FDI), signaling a stable and predictable environment for foreign investors. India's FDI restrictiveness index has remained steady during this period, indicating a consistent and reliable FDI regime. In contrast, China has strategically reduced its index from 0.083 to 0.071, underscoring a deliberate effort to attract increased foreign investments while maintaining necessary regulatory oversight. These nuanced adjustments underscore these nations' dedication to striking a balance between promoting FDI and implementing essential regulatory measures to nurture sustainable economic growth.

FDI: A state perspective

Maharashtra, anchored by the economic powerhouse Mumbai, enjoys a robust financial sector, strategic infrastructure development, and proximity to a substantial consumer market. Karnataka's allure lies in its role as a technology and innovation hub, fostered by a skilled workforce, favorable policies, and a focus on research and development. Gujarat stands out with investments in critical infrastructure, including ports and industrial zones, coupled with business-friendly policies. Delhi, as the national capital, boasts a well-established services sector and a vast consumer base, contributing significantly to its FDI inflow. Tamil Nadu concentrates on automotive and electronics industries, featuring industrial clusters and skilled labour, supported by favorable policies. Haryana, in close proximity to Delhi, thrives on industrial and commercial zones, capitalizing on infrastructural development. Telangana's equity inflow is underpinned by its robust IT and pharmaceutical sectors, facilitated by a conducive business ecosystem. Rajasthan's unique strengths include tourism and mineral resources, while West Bengal's evolving business climate and growing sectors like IT and manufacturing attract investor attention. Each state's distinctive advantages make them attractive to foreign investors, contributing to their noteworthy share of FDI equity inflow.

Figure 9: FDI distribution by state



Note: FDI data for states is from October 2019 to December 2022
Source: Department for Promotion of Industry and Internal Trade

Driving Forces and Challenges in India's Regional Manufacturing Landscape

The contribution of various states to manufacturing GVA throws up an image of regional imbalance. At the forefront of India's manufacturing are states like Gujarat, Maharashtra, and Tamil Nadu, collectively contributing about 40% of the country's manufacturing GVA.

Gujarat's ascendancy in manufacturing is the result of a multifaceted strategy. The state has made strategic investments in infrastructure development, boasting world-class ports and well-planned industrial parks. These facilities provide a solid foundation for industries to thrive. Furthermore, Gujarat's business-friendly environment, characterized by robust policies and ease of doing business, has attracted a diverse array of industries, ranging from automobiles to electronics to petrochemicals and textiles. This diversity not only bolsters the state's economic resilience but also amplifies its attractiveness to investors. Gujarat's logistical advantages, stemming from its excellent connectivity by road, rail, and sea, further cement its position as a manufacturing leader.

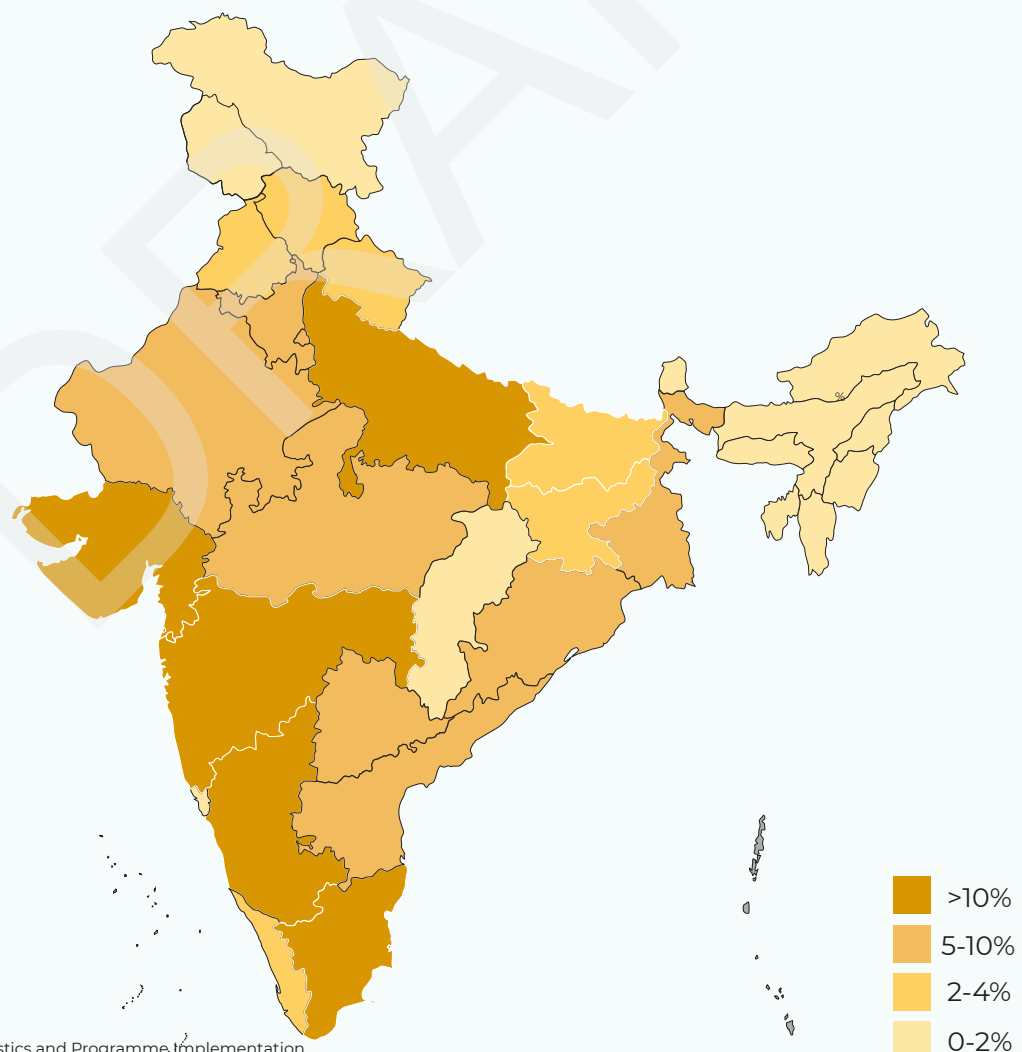
The strategic location, international connectivity and proximity to a large consuming market of Mumbai and Pune have led to Maharashtra accounting for the second largest share of Manufacturing GVA. Maharashtra's extensive infrastructure network, including airports, ports, and industrial zones, significantly contributes to the growth of the manufacturing sector. This infrastructure ensures efficient transportation of raw materials and finished products, enhancing the state's manufacturing GVA.

Tamil Nadu's heft in manufacturing is primarily due to its thriving automotive and electronics sectors. The state's reputation as an ideal investment destination stems from its skilled labour force and supportive policies. Tamil Nadu's proactive approach includes investments in industrial clusters, further enhancing its appeal to global manufacturers. The synergy between these clusters fosters innovation, efficiency, and competitiveness in the manufacturing processes.

States such as Uttar Pradesh, Bihar, West Bengal, Madhya Pradesh and Rajasthan put together contributed 19% to India's manufacturing GVA. Despite a robust workforce, these states have yet to fully realise their manufacturing potential. However, steps have been taken to improve the share of manufacturing in the states GSDP by various policies implemented by the respective state governments.

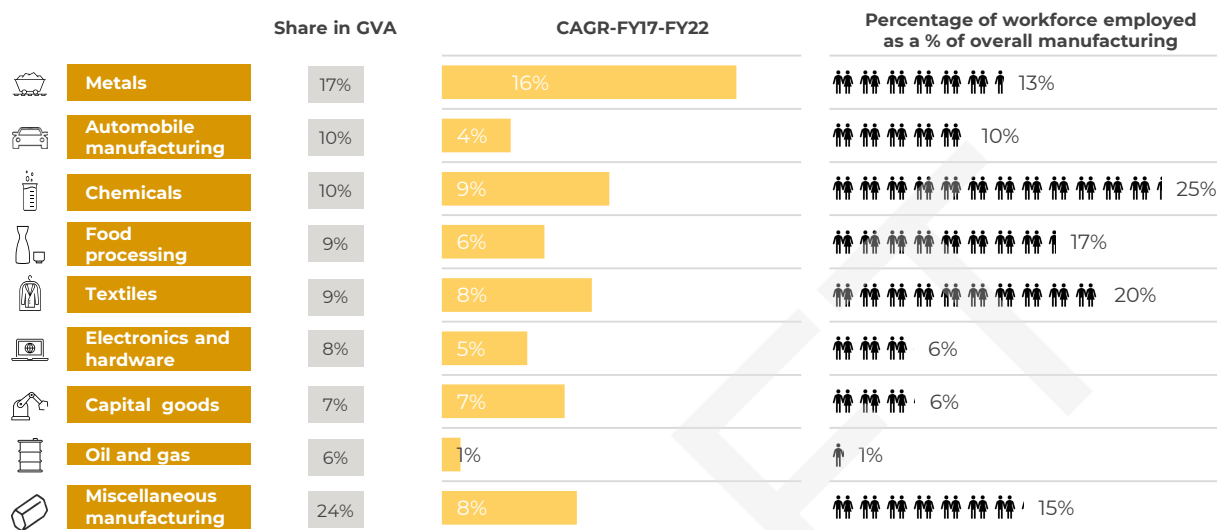
The Northeastern and hilly states put together accounted for about 7% of Manufacturing Gross Value Added (GVA) in India. This relatively low share can be attributed to their unique circumstances. Their rugged terrain and challenging geographical features often pose logistical difficulties for large-scale manufacturing operations. Additionally, these regions prioritize alternative sectors such as tourism, agriculture, and cultural preservation, capitalizing on their natural beauty, cultural heritage, and ecological significance. Furthermore, limited industrial infrastructure and market access may restrict the growth of manufacturing in these areas. Consequently, their economic focus aligns more closely with their distinct strengths and conservation efforts, resulting in a modest contribution to manufacturing GVA.

Figure 10: Manufacturing GVA by state



Top six sectors account for two-thirds of manufacturing GVA

Figure 11: Manufacturing GVA by sector



Source: Ministry of Statistics and Programme Implementation

The metals sector holds a central and indispensable position within India's manufacturing landscape, making substantial contributions to the Gross Value Added (GVA) of the nation. This influence stems from the persistent demand for metals both ferrous and non-ferrous, primarily driven by the construction, infrastructure, and manufacturing industries. As India experiences a continuous wave of urbanization, invests in its infrastructure buildout and rising local consumption and exports, the metals sector continues to stand out as a crucial player, playing a pivotal role in propelling the country's economic growth. The sector's robust performance and adaptability to meet the evolving demands of various industries underscore its significance as a cornerstone of India's industrial and economic development.

Automobile manufacturing has emerged as a cornerstone of India's manufacturing sector, contributing substantially to the economy. The automobile manufacturing sector broadly divided into Original Equipment Manufacturers (OEM's) and automobile component manufacturers. OEM's can further be classified into two-wheeler manufacturers, where India is the largest manufacturer in the world. India ranked 3rd for Passenger vehicle production and 7th for commercial vehicle production in calendar year 2022. The demand for automobiles, fuelled by a burgeoning middle class and increased disposable income, has propelled this sector. With a global transition underway towards alternate mobility, India has an opportunity to emerge as the leader in the transition to electric vehicles and embed itself into global value chains.

The chemicals sector in India is third in the share of manufacturing Gross Value Added (GVA). The chemicals industry serves various segments, encompassing pharmaceuticals, petrochemicals, speciality and agrochemicals. This sector's significance lies in its ability to cater to both domestic demand and global export opportunities, demonstrating its resilience and pivotal role in the manufacturing value chain. The chemical industry not only fuels the growth of ancillary sectors but also plays a crucial role in supporting key industries like agriculture, healthcare, and

manufacturing. The continuous evolution of this sector, marked by technological advancements and a commitment to sustainability, further underscores its importance in driving economic development, innovation, and meeting the intricate needs of a rapidly evolving global market. As India continues to strengthen its foothold in the chemicals domain, the sector is poised to remain a linchpin in the country's manufacturing landscape.

Food processing is integral to India's manufacturing GVA due to the country's agrarian base and a large population dependent on the food industry. To improve India's agriculture output and feed an ever-growing population, the importance of the food processing industry cannot be overstated. As income levels rise, there's a noticeable shift in consumer preferences toward processed and packaged food items, driving the growth of this sector. Moreover, the focus on food safety and quality standards enhances the role of food processing in the manufacturing sector.

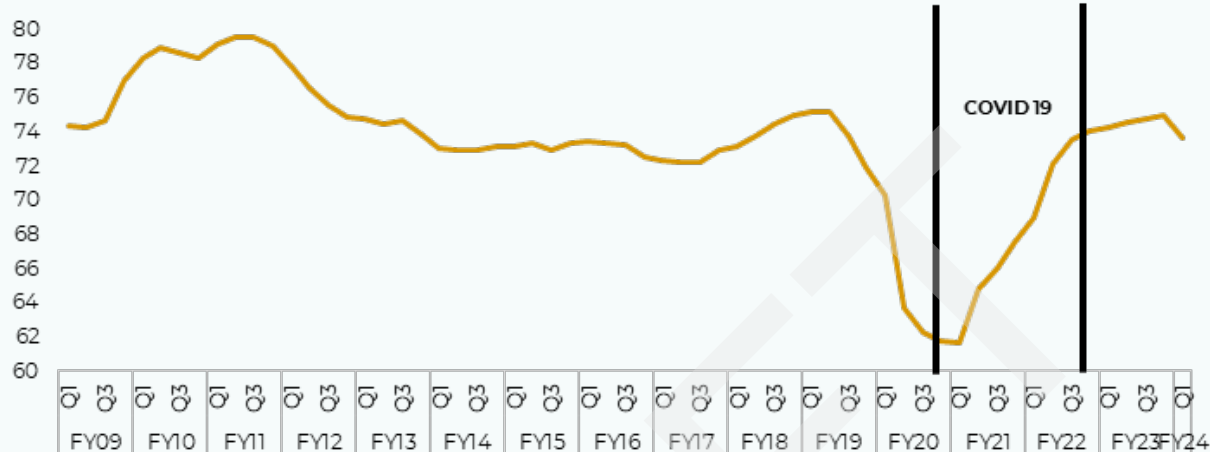
Textiles stand as a cornerstone in the success story of India's manufacturing landscape, making a substantial contribution of approximately 9% to the manufacturing Gross Value Added (GVA) and employment in the country. The Indian textile industry is one of the largest in the world with a large raw material base and manufacturing strength across the value chain. The textile sector in India is expansive, covering a diverse array of products that includes apparel, home textiles, and technical textiles. India's historical affinity with textiles, coupled with its proficient and skilled workforce, positions the nation as a significant player in the global textile market. The sector not only plays a pivotal role in bolstering the national economy but also acts as a critical source of employment generation, particularly in rural areas where traditional craftsmanship thrives. The versatility of the textile industry, spanning from traditional handloom products to technologically advanced technical textiles, underscores its resilience and adaptability. As India continues to harness its rich tradition in textiles, the sector remains a driving force behind the country's manufacturing prowess, fostering economic growth, innovation, and sustainable employment opportunities.

These top six sectors collectively account for two-thirds of India's manufacturing GVA, reflecting their enduring significance in the country's economic landscape. The interplay of domestic demand, global competitiveness, and strategic investments has propelled these sectors to the forefront of India's manufacturing prowess.

Capacity Utilization in Indian Manufacturing sector

Manufacturing firms in India maintained a utilization level in the range of 70-75% to balance flexibility, cost efficiency, and strategic considerations while avoiding overproduction and adapting to market uncertainties. Following the global financial crisis of 2008, the manufacturing sector thrived under the economic stimulus provided worldwide, reaching a peak utilization level of 80% as firms aimed to meet increased production requirements driven by the stimulus. Subsequently, as firms added extra capacities in fiscals 2012 and 2013 and demand softened, capacity utilization dropped back to the normal range of 70-75% beset by hiccups along the way in the form of the Taper Tantrum, Demonetisation and Implementation of GST.

Figure 12: Capacity utilisation in the manufacturing sector



Source: RBI, OBICUS Survey on the Manufacturing sector

During the onset of the pandemic in the last quarter of fiscal 2020, the utilization levels of manufacturing firms in India plummeted to an all-time low of 62% due to widespread disruptions in supply chains, workforce shortages, and government-imposed lockdowns that limited production capabilities. Reduced consumer spending and market uncertainties led many businesses to scale back operations in line with decreased demand. As vaccination efforts progressed and public health conditions improved, governments eased restrictions, enabling businesses to operate more freely. With increased consumer confidence, uptick in the economy and actualization of pent-up demand, industries saw a resurgence in orders and production, contributing to the restoration of utilization levels to 75% as of Q4FY23. Companies also adapted to the evolving situation by implementing more resilient supply chain strategies and adjusting their operations to align with the post-pandemic economic landscape. A number of supply side steps implemented by the government in the forms of schemes and initiatives coupled with the infrastructure buildout in progress and the opportunities arising from India becoming a part of global value chains amid supply chain diversification and rising exports would aid in improving capacity utilisations across sectors giving a boost to capacity utilisations the manufacturing sector is set for healthy demand.

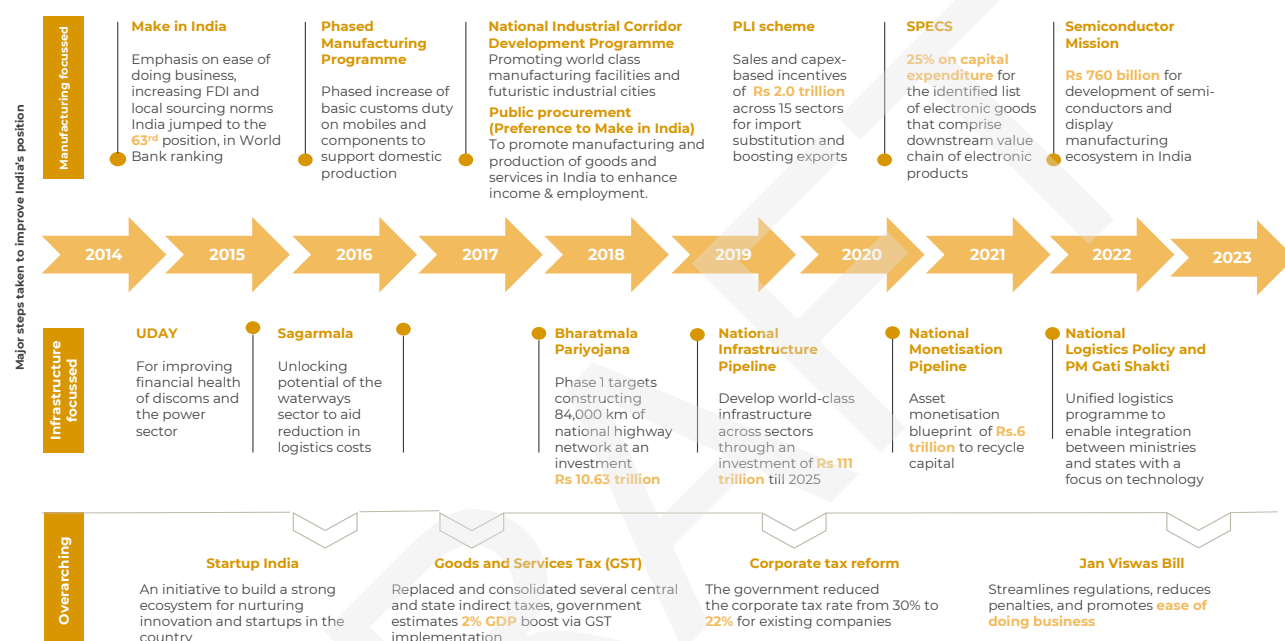
Government Initiatives

Reform leg-up for manufacturing growth

The Indian manufacturing and allied sectors have seen a host of reforms to aid in improving the competitiveness. Reforms can be categorised in three major categories – manufacturing focused, Infrastructure focussed and Overarching reforms. Overarching reforms such as Startup India, Goods and Service Tax have aided in formalisation of the economy and also attracting capital. Infrastructure focused reforms have helped in building out infrastructure to aid in bringing

down power cost, improving connectivity via roads, ports, airports aiding reduction in logistics cost. Manufacturing reforms have helped in seeding new-age sectors while providing a boost to existing manufacturing sectors while helping in employment generation and catering to the export market.

Figure 13: Government initiatives for the manufacturing sector



Make in India: Launched in 2014, Make in India promotes domestic manufacturing and attracts foreign investments across 25 sectors, including automobiles, chemicals, electronics, and textiles. The initiative aims to generate employment, enhance skill development, and increase the manufacturing sector's contribution to India's GDP. Notably, FDI equity inflow in manufacturing increased by 57% from 2014 to 2022, reflecting the campaign's impact on foreign investments in the sector. The campaign focuses on creating a robust industrial ecosystem, fostering economic growth, and positioning India as a global manufacturing hub.

UDAY: Ujwal DISCOM Assurance Yojana (UDAY) was launched with an overall aim of operational and financial turnaround of State-owned Distribution Utilities (DISCOMs) through efficiency improvements and financial restructuring in Generation, Transmission and Distribution Sectors. Salient features of the scheme included providing permanent solutions to legacy debts with state governments taking over 75% of DISCOM debt as on 30th September 2015. DISCOM debt not taken over by the state shall be converted by the banks/FI into loans or bonds with interest rate not more than the bank's base rate of 0.1%. Alternately, this debt may be fully or partly issued by the DISCOM as State guaranteed DISCOM bonds at the prevailing market rates which shall be equal to or less than bank base rate plus 0.1%. The scheme focuses on achieving operational

targets of pan-India AT&C reduction from 22% to 15% by fiscal 2019 along with reduction of the gap between the average cost of supply (ACS) and average revenue realised (ARR). As of fiscal 2020, AT&C losses have reduced to 20.73% while ACS-ARR gap has reduced to Rs. 0.50 per unit from Rs. 0.55 per unit in fiscal 2016. Since its launch, 32 states and union territories have joined the scheme.

Phased Manufacturing Programme: The Phased Manufacturing Programme (PMP) is a government initiative aimed at encouraging domestic manufacturing of mobile phones and related components in India. The program is implemented in a phased manner, with the goal of increasing domestic value addition and reducing imports over time.

Sagarmala: The Sagarmala Programme, led by the Ministry of Ports, Shipping, and Waterways, is a strategic initiative aimed at revolutionizing India's maritime sector. Focused on socio-economic development, it promotes efficient and sustainable transportation through initiatives such as Ro-Ro ferry services. The program, backed by financial support of ~Rs. 1,900 Crore prioritizes urban water transportation projects. Its business-centric approach, as evidenced by model agreements and guidelines, seeks to streamline operations, enhance safety, and attract domestic and international investments for transformative growth in the maritime industry.

National Industrial Corridor Development Programme: The National Industrial Corridor Development Corporation (NICDC), operating under the flagship National Industrial Corridor program, has strategically identified and designated 11 industrial corridors in four distinct phases. The primary objective is to establish state-of-the-art, world-class infrastructure aimed at fostering local commerce, boosting investment, generating employment, and achieving sustainable development. The overarching vision is to position India as a global hub for manufacturing and investment, leveraging a high-capacity transportation network, developing "smart cities" as new industrial hubs, and integrating cutting-edge technologies across sectors to catalyse employment opportunities and economic growth, thereby contributing to comprehensive socio-economic development.

Public Procurement: The Public Procurement (Preference to Make in India) Order 2017 is a governmental policy aimed at fostering domestic manufacturing. It introduces preferences and incentives in the procurement process for goods and services produced within India. By prioritizing locally made products, the order intends to stimulate economic growth, generate employment, and support the country's manufacturing sector. This initiative aligns with the broader "Make in India" campaign, encouraging both domestic and international entities to contribute to India's self-sufficiency and economic development through increased production within the nation.

Bharatmala Pariyojana: It is a strategic venture led by the Ministry of Road Transport and Highways, aims to optimize India's national transportation network. In its initial phase, the project focuses on constructing new corridors and establishing NH linkages to bolster economic growth. With an approved budget of Rs. 5.35 Lakh Crore, it covers 24,800 km of diverse roads and 10,000 km of residual NHDP projects. The initiative prioritizes connectivity and economic development, with 23,500 km awarded and 11,400 km completed, emphasizing the significance of Economic Corridors (12,002 km) and Coastal & Port Connectivity roads (1,079 km) in strategically investing in the nation's transport infrastructure.

National Infrastructure Pipeline: The National Infrastructure pipeline (NIP) outlines an investment of Rs. 111 lakh crores across fiscals 2020 to 2025 in infrastructure spanning a number of sectors such as Roads, Railways, Power, Ports, Airports, Urban, Digital, Irrigation, Rural, Agriculture & Food processing, Social and Industrial. It outlines creating new and upgrading existing infrastructure

which aids in India's competitiveness and is critical for the success of the Make in India and PLI programme as manufacturing competitiveness critically depends on infrastructure. Improved infrastructure capacities also create efficiency gains through improved logistics and networks, which would improve the competitiveness of the economy.

Production Linked Incentive Scheme: The Production Linked Incentive (PLI) Scheme, a key government initiative, aims to bolster India's manufacturing capabilities and exports across 14 strategic sectors. With an outlay of Rs. 1.97 lakh crore, the scheme focuses on critical areas such as electronics, pharmaceuticals, automobiles, and textiles. By attracting investments and promoting cutting-edge technology, it seeks to enhance efficiency, achieve economies of scale, and make Indian businesses globally competitive. The scheme, operational in various stages, holds the potential to significantly boost production, employment, and economic growth over the next five years, transforming India into a self-reliant and globally competitive manufacturing hub.

Semiconductor Mission: The Semiconductor Mission, with its comprehensive strategy and financial support, aligns with the government's broader vision of reducing import reliance, promoting indigenous semiconductor fabrication, and advancing research and development. By nurturing semiconductor design capabilities and fostering self-sufficiency, the mission positions India as a key player in the global semiconductor market. This initiative, backed by INR 76,000 crore, serves as a catalyst for innovation, economic advancement, and job creation, reinforcing the nation's commitment to technological excellence and leadership in the electronics industry.

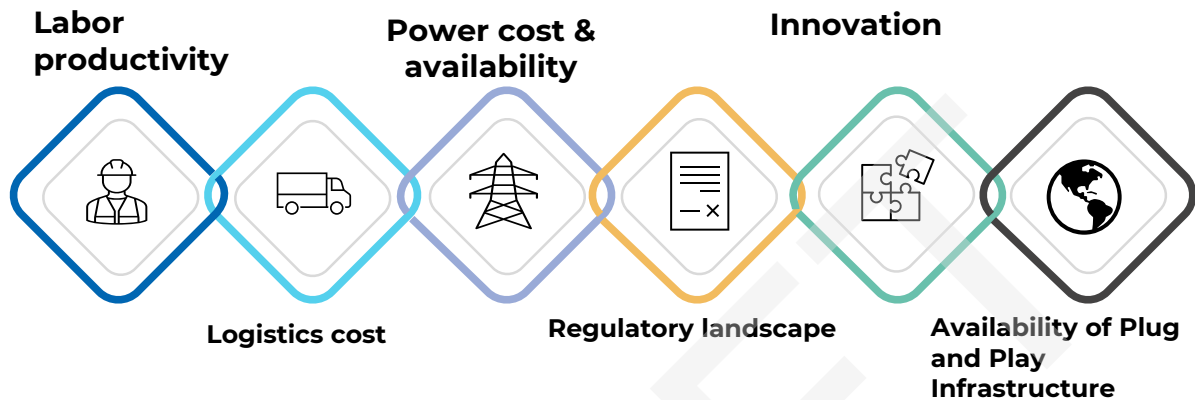
National Monetisation Pipeline: The National Monetisation Pipeline is a government initiative to unlock value from existing public sector assets. This strategic approach involves leasing operational infrastructure assets like roads, airports, and railways to the private sector, generating revenue and enhancing efficiency. Aimed at attracting Rs 6.0 lakh crores in private sector investment from FY 2022 to FY 2025 for infrastructure creation, the program emphasizes brownfield de-risked assets. Focused on revenue rights rather than ownership, assets are returned to the government at the end of the transaction life, fostering sustainable infrastructure financing and economic growth.

PM-MITRA: The PM MITRA scheme entails establishing seven Mega Integrated Textile Region and Apparel (PM MITRA) parks through public-private partnerships to catalyse the growth of the textile sector. Aligned with the Azadi Ka Amrit Mahotsav, it envisions an investment of approximately Rs. 70,000 crores, fostering a comprehensive textile value chain from farm to fashion. Prioritizing world-class infrastructure to attract cutting-edge technology and foreign direct investment, the scheme seeks to position India as a global textile manufacturing and export hub. With a transparent selection process, financial support, and collaboration between central and state governments, it aims to generate 20 lakhs jobs.

National Logistics Policy and PM-Gatishakti: The National Logistics Policy and PM-Gatishakti initiative focus on streamlining and optimizing logistics and infrastructure networks in India. The policies aim to enhance connectivity, reduce inefficiencies, and promote a seamless movement of goods. National Logistics Policy emphasizes creating a single-window e-logistics marketplace, fostering transparency and efficiency. PM-Gatishakti involves developing a holistic infrastructure master plan, integrating various modes of transport, and strengthening connectivity to boost economic growth and competitiveness. Both initiatives prioritize a robust and integrated logistics framework for sustained development.

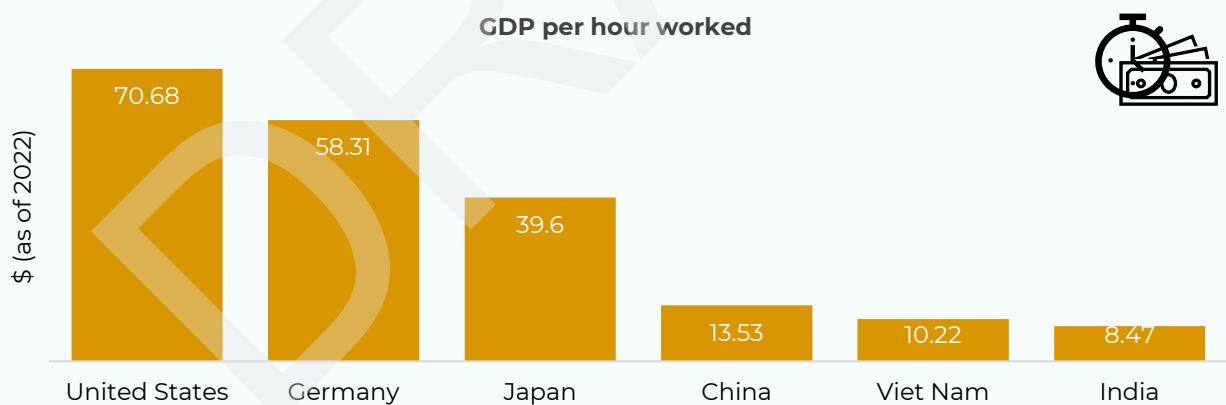
Challenges

Figure 14: Challenges in Indian Manufacturing



1. Labour cost competitive, but productivity gap plays spoilsport

Figure 15: GDP Per Hour Worked:

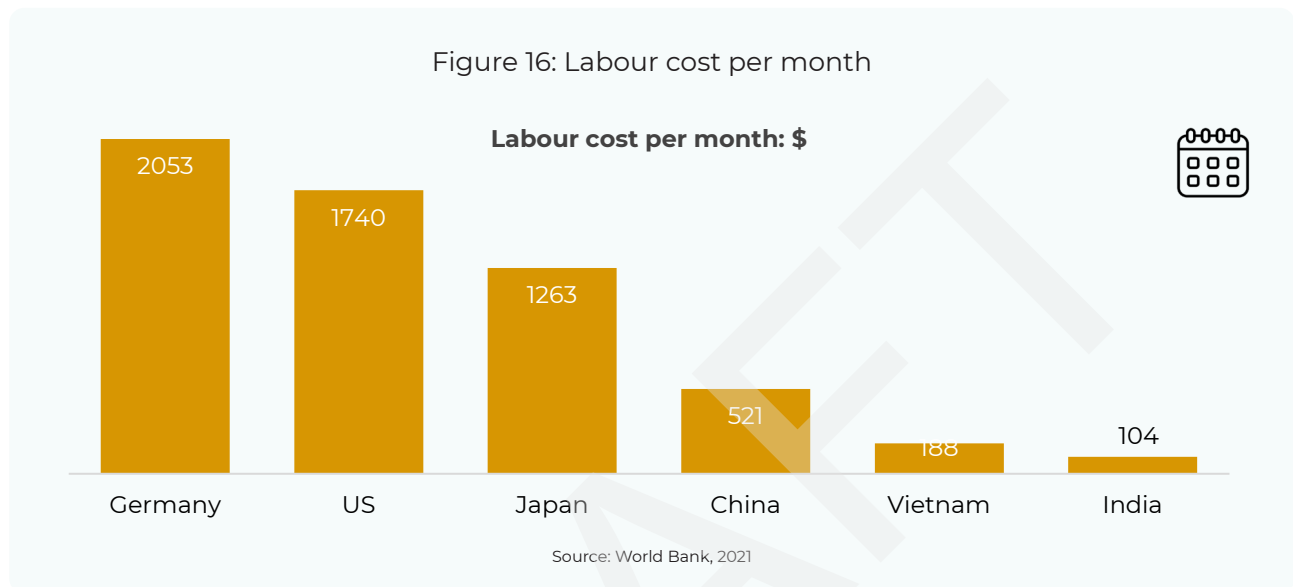


Source: Organisation for Economic Co-operation and Development

GDP per hour worked is a measure of labour productivity. It measures how efficiently labour input is combined with other factors of production and used in the production process. Labour input is defined as total hours worked of all persons engaged in production

GDP per hour worked varies significantly among these nations, reflecting diverse economic landscapes and productivity levels. The United States leads with its advanced technology adoption and innovation, resulting in the highest GDP per hour worked. Germany closely follows, known for its precision engineering and manufacturing prowess. South Korea's strong industrial base, particularly in electronics and automotive sectors, contributes to its productivity. Japan excels in technology-driven industries. In contrast, China, with a vast and varied workforce, experiences

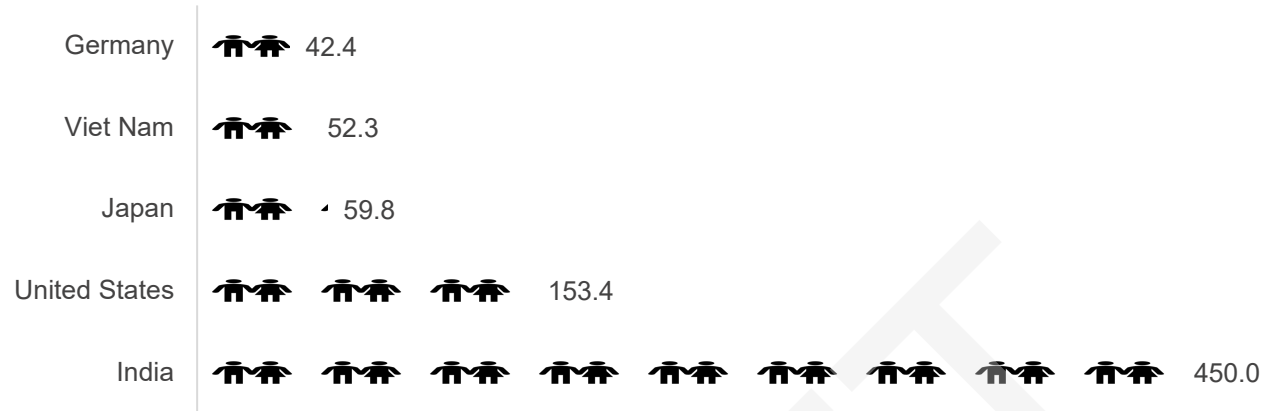
disparities in productivity due to urban-rural distinctions and technology utilization. Vietnam focuses on export-oriented manufacturing, while India faces challenges related to its economic structure and skill disparities. These differences emphasize the role of technology, industrial strength, and economic composition in shaping labor productivity globally.



Labor costs exhibit significant variations across different countries, driven by an intricate interplay of diverse economic, social, and regulatory factors. These disparities reflect the unique characteristics and dynamics of each nation's labour market.

In the United States, labour costs are relatively high due to a robust framework of labour regulations, ensuring workers' rights and a generally higher cost of living. Germany also boasts elevated labour costs, primarily stemming from strong labour unions and an extensive social welfare system that supports employees. Japan maintains its relatively higher labour costs, reflecting a well-established infrastructure, strict labour laws, and a workforce with advanced skills. In contrast, countries like China, Vietnam, and India experience lower labour costs attributed to factors such as a large labour force, varying economic development stages, and more competitive wage structures. These disparities underscore the complexity of labour cost determination, shaped by factors unique to each country's economic landscape. Businesses must carefully consider these nuances when making strategic decisions about their operations and investments on a global scale.

Figure 17: Labour Force

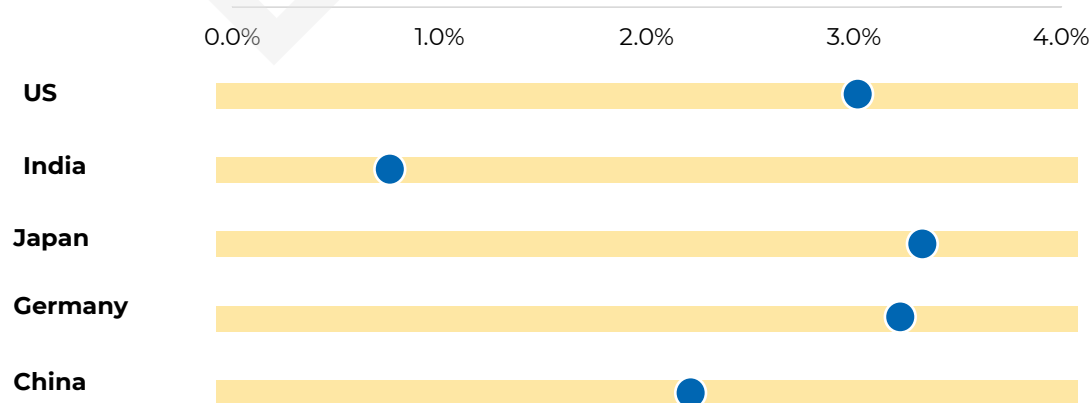


Source: World Bank, 2022

Labor force comprises people ages 15 and older who supply labor for the production of goods and services during a specified period. It includes people who are currently employed and people who are unemployed but seeking work as well as first-time jobseekers. Not everyone who works is included.

India boasts the world’s largest labor force, providing a significant advantage for industries and businesses seeking a diverse and skilled workforce. The United States benefits from a well-trained and innovative workforce that drives economic growth and technological advancements. Japan’s labor force contributes to its advanced manufacturing and technological sectors, fostering competitiveness on the global stage. Vietnam’s sizable workforce is a crucial asset, particularly in manufacturing and export-oriented industries. Germany is renowned for its highly skilled labor force, supporting its position as an industrial powerhouse. South Korea’s workforce adds to its technological prowess and manufacturing excellence. These labor force advantages, whether in scale, skills, or specialization, play a pivotal role in shaping these countries’ economic landscapes and enhancing their global competitiveness.

Figure 18: Employee Expense as a % of total expense



Source: World Bank, 2021

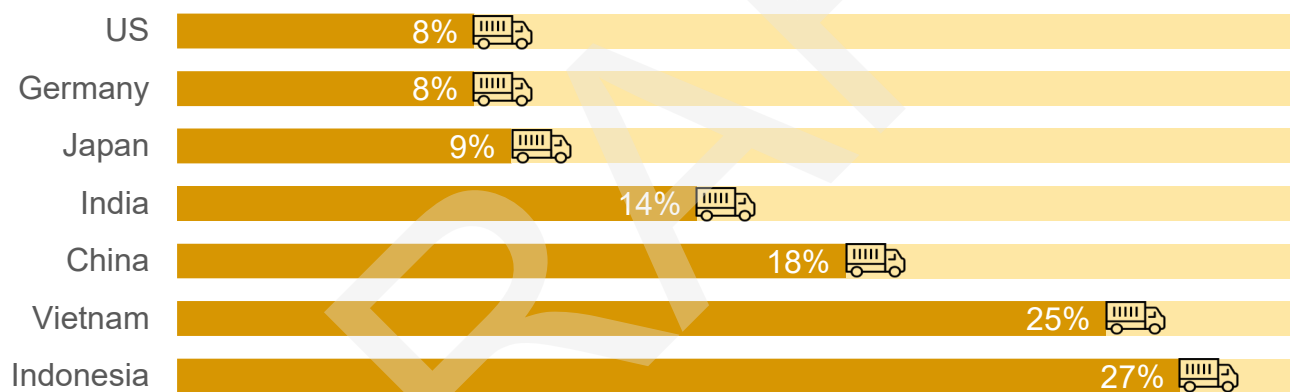
South Korea, the United States, and India showcase relatively high percentages, indicating substantial labour costs that underscore their reliance on labour-intensive industries. These nations typically have large workforces engaged in diverse sectors, contributing significantly to overall expenses. This can be advantageous as it reflects a thriving job market, robust consumer spending, and, in some cases, higher living standards.

On the flip side, countries like Japan and Germany exhibit lower percentages, suggesting a potentially lower dependency on labour-intensive sectors and greater automation or capital-intensive industries.

Despite India possessing a large labour force and among the lowest labour cost, peer nations fare better on account of increased automation and higher productivity.

2. Logistics landscape: Catching up with developed economies

Figure 19: Logistics cost as a % of GDP

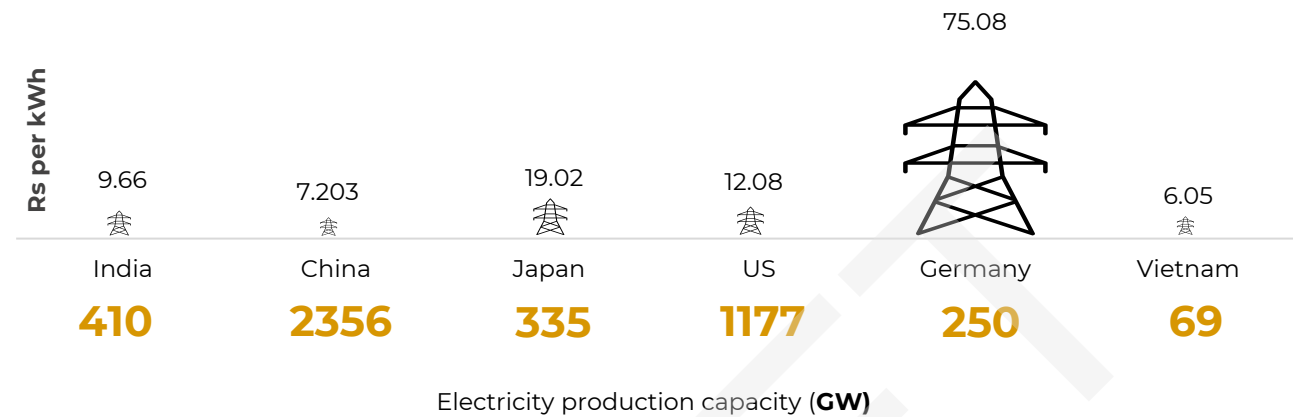


Source: World Bank

As India's logistics costs stand at 14% of its GDP, it's evident that the nation is in the midst of an evolution to catch up with developed economies. While not the highest in the world, this percentage highlights room for optimization. Compared to developed counterparts like the United States and Germany, where logistics costs hover around 8%, India's logistics ecosystem presents both challenges and opportunities. To bridge this gap effectively, India must focus on streamlining operations, reducing costs, enhancing infrastructure, and fostering innovation within its logistics sector. This transformation is pivotal for India's economic competitiveness, trade facilitation, and its journey toward becoming a global logistics and manufacturing hub.

3. India's power cost: A challenge to competitiveness

Figure 20: Benchmarking India's power cost



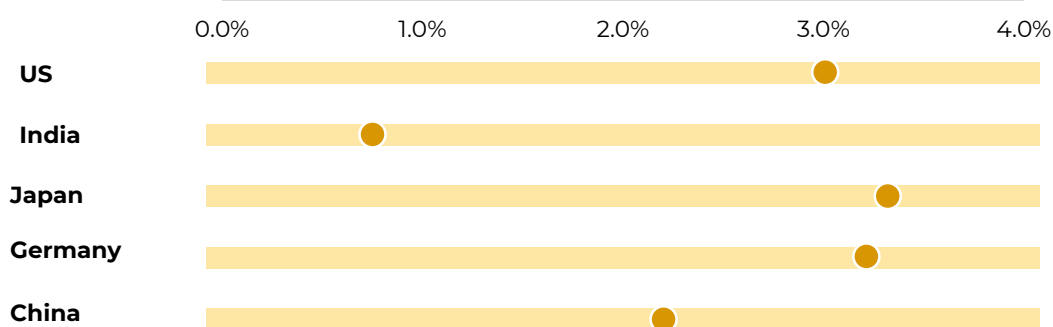
Sources: Global Petrol Prices, Ministry of Power, Power Finance Corp
Note: All data pertains to CY2022
The installed capacity for India as per Ministry of Power

India's relatively high-power cost of 9.66 Rs/kWh compared to countries like China (7.203 Rs/kWh), Vietnam (6.05 Rs/kWh), and even the United States (12.08 Rs/kWh) places it at a disadvantage from a manufacturing standpoint. This is significant because manufacturing heavily relies on affordable and consistent electricity to power machinery and operations. Despite India's substantial electricity production capacity of 410 gigawatts, the manufacturing sector still faces challenges related to grid reliability and distribution. These challenges can lead to occasional power shortages and interruptions, causing downtime and increasing production costs. In contrast, countries with lower power costs and robust electricity infrastructure, like China and the United States, offer manufacturing industries a competitive edge by reducing operational expenses.

To enhance its manufacturing competitiveness, India needs to address its power cost issues and invest in grid reliability, ensuring a steady and affordable power supply for the sector. Such improvements are pivotal for attracting investments, fostering growth, and maintaining a competitive edge in the global manufacturing landscape.

4. Innovation

Figure 21: Research and development expenditure (% of GDP)



Source: World Bank
Manufacturing in India

The R&D expenditure as a percentage of GDP for these countries reveals their distinct approaches to innovation and technological advancement. China's allocation of 2.1% of GDP reflects its ambition to become a global tech leader, driving advancements in various sectors. Germany, with 3.1%, maintains its reputation as an innovation powerhouse, especially in engineering and manufacturing. India's 0.7% signifies its growing commitment to research, aiming to boost scientific capabilities and foster innovation-driven growth. Japan's substantial 3.2% investment highlights its dedication to cutting-edge technologies, crucial for its electronics and automotive industries. The United States, allocating 3.0%, continues to lead in innovation, fuelling progress in IT, biotechnology, and aerospace. These percentages underscore each nation's strategic emphasis on R&D, vital for technological progress, economic resilience, and global competitiveness.

5. Availability of plug and play infrastructure

. While progress has been made in several aspects of this ambitious initiative, the real challenge lies in creating essential social infrastructure. The NICDC, initially conceived as the Delhi Mumbai Industrial Corridor (DMIC), is still in the process of completion. Formidable hurdles, particularly in land acquisition and aggregation, persist. Overcoming these challenges is crucial for the successful establishment of the corridors, followed by the intricate tasks of selling these land parcels and attracting industrial units. Addressing these challenges will be pivotal in realizing the full potential of the NICDC and ensuring its transformative impact on India's industrial and economic landscape.

Way forward



Ease of Doing Business

India's rise in the now-discontinued World Bank's Ease of Doing Business Ranking, from 142nd in 2014 to 63rd in 2019, suggests a favourable environment for investments and foreign capital, positioning India as a hub for investment and industrial growth contributing to its growth. Simultaneously, it plays a role in attracting foreign capital, contributing to the nation's economic growth. A continued policy focus on improving the ease of doing business is imperative, particularly addressing areas where India lags its peers and offers room for improvement.



Competition

acts as a dynamic force propelling the progression of India's manufacturing sector, a principle underscored by the recently introduced Competition (Amendment) Bill, 2022. This legislation takes aim at curbing anti-competitive practices, streamlining merger approvals, and introducing deal value thresholds. Collectively, these measures contribute to a more competitive market landscape, fostering efficiency, innovation, and an environment conducive to business growth. The outcome is a strengthened manufacturing sector, poised for enhanced global competitiveness. Need based Adjustments to the Bill could be pursued as needed, ensuring its continuous refinement and effectiveness in promoting a thriving business ecosystem



Taxation

A well-structured tax code is easy for taxpayers to comply with and can promote economic development while raising sufficient revenue for a government's priorities. In contrast, poorly structured tax systems can be costly, distort economic decision-making, and harm domestic economies. A number of steps across direct taxation, indirect taxation, processes and tax settlements have been implemented to streamline India's tax structure and processes and make it competitive with peers. Corporate tax rates have been reduced to 20% and 15% excluding surcharge making them competitive with peers. Average GST monthly collections have doubled between fiscal 2018 and 1st half of fiscal 2024. A few tweaks have been carried out in the GST regime based on feedback received as well as consultations with stakeholders and such a process can be pursued to ensure robust collections.



Enforcing contracts and Dispute resolution

India ranked 163rd globally in the Ease of Doing Business 2020 on this aspect. The prolonged legal processes and delays in resolving contractual disputes undermine business confidence, increasing uncertainty for investors. To address this, the Commercial Courts Act, 2015 was introduced, emphasizing the importance of proper implementation and continuous adaptation to the evolving landscape. Leveraging digital platforms for case management and promoting alternative dispute resolution mechanisms supported by legislative advancements like the Arbitration and Conciliation Act, 1996, the India International Arbitration Centre Act, 2019, and the Mediation Bill, 2021, signal India's commitment to a global standard legal framework, fostering a conducive environment for investments and sustainable business development. Effective implementation of the arbitration act is vital for attracting FDIs in this current scenario of global supply chain diversification.



Quality Control and Inspection

The fortification of the Indian manufacturing industry demands a dedicated emphasis on elevating quality control and inspection practices. This entails a comprehensive strategy that involves aligning with global quality benchmarks, integrating advanced monitoring technologies, fostering the continuous upskilling of the workforce, securing pertinent certifications such as ISO, conducting thorough supplier audits, implementing robust feedback mechanisms, and ensuring meticulous digital record-keeping. The DPIIT has taken concrete steps to establish a comprehensive quality control regime in India. One of the key measures undertaken by DPIIT is the development of QCOs with the aim to curb the import of sub-standard products, prevent unfair trade practices, and ensure the safety and well-being of consumers, as well as the environment. The implementation of the QCO is envisaged to improve manufacturing quality standards and enhance the brand value of 'Make in India' products. In collaboration with the Bureau of Indian Standards (BIS), DPIIT has initiated the development of 64 new QCOs, covering 317 product standards. Quality Control Orders (QCOs) should be broadened to encompass a more extensive range of products, while the application standards such as ISI marks and BEE ratings should be extended across various product categories, serving as indicators of commendable product quality.



National Single Window Clearance

India's National Single Window Clearance system is a transformative initiative streamlining bureaucratic processes, expediting project approvals and permits. This approach simplifies interactions with government authorities, fostering an investor-friendly environment and supporting industrial growth, enhancing India's appeal as a business destination. Notably, the system's expansion is underway, with 16 states/UTs and 26 central government departments already onboarded, including Andhra Pradesh, Bihar, Goa, Gujarat, Himachal Pradesh, Jammu & Kashmir, and Karnataka. Extending this initiative to all states, UTs, and departments will make it widely applicable.



Labor Reforms

India's labor law reforms have brought flexibility and efficiency to the manufacturing sector. Key measures such as the Industrial Relations Code, Occupational Safety, Health, and Working Conditions Code, Social Security Code, and Code on Wages streamline labor laws. Complemented by policies like "Skill India," "Stand Up India," and "Pradhan Mantri Rojgar Protsahan Yojana," these reforms create a conducive environment for industrial growth.



Skill Development

With a burgeoning population and a vast demographic dividend, India should capitalise on the opportunity by undertaking skill development of its youth. This concerted effort aims to provide job-ready skills to a vast youth population, revitalizing and strengthening the manufacturing sector while bridging skill gaps to boost employability. Skill development enhances workforce productivity and global competitiveness in Indian manufacturing. Central government policies such as Pradhan Mantri Kaushal Vikas Yojana, National Skill Development Mission, and National Apprenticeship Promotion Scheme align with the "Skill India" initiative launched in 2015 are some steps taken in the direction.



Infra- structure Investments

The Government's budget allocation for capital expenditure (capex) has shown steady growth, with a 27% Compound Annual Growth Rate (CAGR) between fiscal years 2019 and 2024. These investments play a significant role in strengthening the manufacturing sector by improving critical infrastructure. Complementary initiatives, such as Sagarmala, Dedicated Freight Corridors (DFCs), and Bharatmala, further enhance these efforts. Sagarmala focuses on port modernization and coastal connectivity, while DFCs prioritize efficient freight movement, and Bharatmala improves road connectivity. The National Infrastructure Pipeline (NIP), National Logistics Policy (NLP), and "GatiShakti" are also set to streamline logistics and reduce costs, benefiting manufacturing efficiency and global competitiveness.



Updating India's Manu- facturing Policy

The National Manufacturing Policy, introduced in 2011, initially outlined targets until 2022. Subsequently, numerous schemes, programs, and initiatives have been implemented to bolster the manufacturing sector, spurred by the emergence of various sunrise sectors and opportunities resulting from supply chain diversification. Recognizing the evolving landscape and the advent of new-age sectors, an overarching policy encompassing a host of schemes, initiatives, strategies, identification of emerging sectors would aid in aligning with the evolving dynamics of the manufacturing sector and ensuring its continued growth and relevance in the changing global economic landscape and aid in meeting the target for a \$35-45 trillion economy by 2047.



Stable custom regime and Free Trade Agreements (FTA's)

India has been able to grow her merchandise exports by 8% annually between fiscals 2018 and 2023. This period entails the general economic slowdown, the Covid-19 pandemic impacted years as well as monetary tightening which makes the achievement even more commendable. A stable custom regime as well as signing of FTAs under discussion with developed nations should provide a further leg up to exports. By streamlining trade processes, reduce trade barriers, and enhance the nation's status as a global manufacturing and export hub.



Reducing India's Logistics cost

India ranks 40th in the Logistic performance index with high logistics cost making India's exports expensive against competing nations. India aims to optimize logistics costs through initiatives like the National Infrastructure Pipeline (NIP) and the National Logistics Policy (NLP) with a stated aim of achieving price parity with competing nations. The NIP prioritizes improved transportation infrastructure, aligning with the goal of reducing inefficiencies. Simultaneously, the NLP focuses on enhancing operational efficiency, streamlining logistics operations. The Dedicated Freight Corridor Corporation (DFCC) plays a pivotal role in elevating freight movement efficiency, complementing the overall logistics strategy. Additionally, the development of Multi-Modal Industry Parks, Container Freight Stations (CFS), and Inland Container Depots (ICD) stations strengthens the logistical landscape. Expansion of port facilities and prioritizing highway quality improvement further contribute to the holistic approach. Digital solutions deployment and regulatory simplification remain central to systematically addressing logistics challenges, fostering competitiveness. This integrated strategy aligns with India's commitment to fortifying its logistics ecosystem, driving economic growth, and establishing the country as a significant global trade player.



Attracting Emerging sector

To position India as a global manufacturing hub, emphasis should be placed on attracting emerging sectors such as semiconductors, solar PV manufacturing, electric vehicles, drones, and wearables. These industries hold significant growth potential and represent the future of manufacturing. Encouraging investments and fostering research and development in these sectors will contribute to India's global competitiveness. By focusing on these key areas, India can align itself with the global trend toward sustainable and technology-driven manufacturing, solidifying its position as a key player in these evolving industries

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